

Web Data Integration
Project Outline

**Pharmaceutical Drug Integration: Linking
US and EU Approval Data**

Gabriel Himmelein (1649181)
Özlem Ölçer (2265825)
Anatole Lobenko (2187322)
Hai
Thu Huyen Trang Trinh (2111810)

October 6, 2026

Submitted to
Data and Web Science Group
Dr. Ralph Peters
Prof. Dr. Christian Bizer
University of Mannheim

1 Brief description of use case

This project aims to build a unified view of approved pharmaceutical drugs in the United States and the European Union, enabling researchers and analysts to query regulatory status, approval history, and product attributes across both markets from a single integrated data set. To achieve this, we combine data from three complementary sources. Drugs@FDA is the FDA’s official database of US-approved drugs. EMA Medicines is the European Medicines Agency’s register of EU-authorised medicines. The openFDA NDC Directory is a supplementary US drug listing with broader product coverage. The central entity is the active substance, the pharmacologically active ingredient in a drug product. Active substances serve as a natural integration key because they persist across different brand names, manufacturers, regulatory jurisdictions, and formulations. The same substance can even carry different names across regions, as with paracetamol in the EU and acetaminophen in the US. Beyond regional synonyms, biosimilars carry manufacturer-specific suffixes that further complicate direct matching. Resolving these naming variations through identity resolution is therefore a central task of the project, ultimately producing one clean record per active substance with consolidated attributes from all three sources.

2 How the datasets fulfill the requirements

2.1 Schema and basic profile of each dataset

Table 1 summarises the three input datasets. Table 2 lists the full attribute sets of Drugs@FDA and EMA Medicines.

Table 1: Overview of the input datasets

Dataset	Entities	Attributes	Data Format
Drugs@FDA	51,864 products	37	TXT/TSV (12 tables)
EMA Medicines	2,746 medicines	39	XLSX
openFDA NDC	138,209 products	10 (selected)	JSON → CSV

Drugs@FDA

Drugs@FDA contains **51,864 product records** across **29,377 applications**. The product table contains only **8,388 distinct drug names** and **3,355 distinct active-ingredient strings**, showing substantial reuse across products. Consequently, **Drug-Name + ActiveIngredient** provides a more appropriate starting point for entity linkage than DrugName alone.

The dataset is strongly imbalanced. Among applications, **78.3% are ANDAs**, **20.1% NDAs**, and **1.7% BLAs**. Among 194,004 submission records, **85.6% are supplemental**, while **99.4%** have submission status AP.

Three attributes exceed 30% missingness: `ApplPublicNotes` (**99.99%**), `SubmissionsPublicNotes` (**99.83%**), and `ReviewPriority` (**41.73%**). The first two are optional regulatory note fields and are therefore populated only in exceptional cases. Missing `ReviewPriority` values cannot be explained by submission type alone: missingness remains high for both original submissions (**49.5%**) and supplements (**40.4%**). Overall, `ReviewPriority` consists of **49.3% Standard**, **5.8% Priority**, **3.2% other codes**, and **41.7% missing values**.

EMA Medicines

The EMA dataset contains **2,746 medicine records** and **39 attributes**, with no duplicate rows. It is predominantly composed of human medicines (**85.6%**). Medicine status is also imbalanced: **68.5% are currently authorised**, **15.4% withdrawn**, and **10.4% represent withdrawn applications**. Fifty medicine names occur more than once, again showing that names alone do not uniquely identify entities.

Twelve EMA attributes exceed 30% missingness. Most of this is **structural rather than erroneous**. Veterinary attributes are inapplicable to human medicines, while refusal, withdrawal, suspension, and rolling-review dates exist only if the respective regulatory event occurred. Two more broadly applicable attributes require closer inspection: `Opinion adopted date` is **46.2% missing**, and `Revision number` is **30.2% missing**.

openFDA NDC Directory

The openFDA National Drug Code (NDC) Directory contains product listing information submitted to the FDA by drug labelers. The dataset was downloaded in JSON format and converted to CSV. It contains 138,209 product records and 7,682 distinct active ingredient entries. The listing includes both FDA-approved and unapproved products. Inclusion in the directory does not imply FDA approval.

Attributes: product NDC, brand and generic names, active ingredients, manufacturer, dosage form, route, marketing start date, marketing category, and application number. Marketing start date is not necessarily the approval date.

Across the 10 selected columns, 113,976 of 1,382,090 values are missing, which is approximately 8.25%, assuming the remaining columns have no missing values. The missing values by column:

- `brand_name`: 21,716 (15.71%)
- `generic_name`: 3 (less than 0.01%)
- `active_ingredients`: 2,521 (1.82%)
- `manufacturer`: 28,052 (20.30%)
- `route`: 23,596 (17.07%)
- `application_number`: 38,088 (27.56%)

Attribute overview

Attributes marked with ** have more than 30% missing values.

Table 2: Attributes of Drugs@FDA and EMA Medicines

Dataset	Source (*)	Format	# Entities	# Attributes	List of attributes (**)
Drugs@FDA	Official U.S. FDA Drugs@FDA Data Files. Downloaded and extracted into 12 relational text tables.	TXT/TSV 12 relational tables	51,864 products 29,377 applications	56 table-columns 37 unique names	ActionTypes_Lookup: ActionTypes_LookupID, ActionTypes_LookupDescription, SupplCategoryLevel1Code, SupplCategoryLevel2Code. ApplicationDocs: ApplicationDocsID, ApplicationDocsTypeID, ApplNo, SubmissionType, SubmissionNo, ApplicationDocsTitle, ApplicationDocsURL, ApplicationDocsDate. Applications: ApplNo, ApplType, ApplPublicNotes **, SponsorName. ApplicationsDocsType_Lookup: ApplicationDocsType_LookupID, ApplicationDocsType_Lookup_Description. Join_Submission_ActionTypes_Lookup: SubmissionType, j_submissionActionTypeID, ApplNo, SubmissionNo, ActionTypes_LookupID. MarketingStatus: MarketingStatusID, ApplNo, ProductNo. MarketingStatus_Lookup: MarketingStatusID, MarketingStatusDescription. Products: ApplNo, ProductNo, Form, Strength, ReferenceDrug, DrugName, ActiveIngredient, ReferenceStandard.

Dataset	Source (*)	Format	# Entities	# Attributes	List of attributes (**)
					SubmissionClass_Lookup: SubmissionClassCodeID, SubmissionClassCode, SubmissionClassCodeDescription. SubmissionPropertyType: ApplNo, SubmissionType, SubmissionNo, SubmissionPropertyTypeCode, SubmissionPropertyTypeID. Submissions: ApplNo, SubmissionClassCodeID, SubmissionType, SubmissionNo, SubmissionStatus, SubmissionStatusDate, SubmissionsPublicNotes **, ReviewPriority **. TE: ApplNo, ProductNo, MarketingStatusID, TECode.

Dataset	Source (*)	Format	# Entities	# Attributes	List of attributes (**)
EMA Medicines	Official European Medicines Agency “Download medicine data” dataset. Imported from the official XLSX file.	XLSX	2,746 medicines	39	Category; Name of medicine; EMA product number; Medicine status; Opinion status ^{**} ; Latest procedure affecting product information; International non-proprietary name (INN) / common name; Active substance; Therapeutic area (MeSH); Species (veterinary) ^{**} ; Patient safety; ATC code (human); ATCvet code (veterinary) ^{**} ; Pharmacotherapeutic group (human); Pharmacotherapeutic group (veterinary) ^{**} ; Therapeutic indication; Accelerated assessment; Additional monitoring; Advanced therapy; Biosimilar; Conditional approval; Exceptional circumstances; Generic; Orphan medicine; PRIME: priority medicine; Marketing authorisation developer / applicant / holder; European Commission decision date; Start of rolling review date ^{**} ; Start of evaluation date ^{**} ; Opinion adopted date ^{**} ; Withdrawal of application date ^{**} ; Marketing authorisation date; Refusal of marketing authorisation date ^{**} ; Withdrawal / expiry / revocation / lapse of marketing authorisation date ^{**} ; Suspension of marketing authorisation date ^{**} ; Revision number ^{**} ; First published date; Last updated date; Medicine URL.

(*) **Source.** Drugs@FDA was obtained from the official FDA Drugs@FDA Data Files download and extracted into its 12 relational tables. EMA medicine data was obtained from the official EMA downloadable medicine-data spreadsheet.

(**) **Missingness.** Attributes marked with ^{**} contain more than 30% missing values. High missingness does not necessarily indicate poor data quality: some attributes are optional,

category-specific, or populated only when a particular regulatory event occurs.

2.2 Integrated schema and overlap with input schemata

Because the entity is the active ingredient, the integrated schema aggregates product-level information from all three sources into one record per ingredient. It comprises ten attributes (Table 3). Eight appear in at least two input schemata, and four appear in all three.

activeIngredientName is the primary matching key, mapped from ActiveIngredient (Drugs@FDA), Active substance (EMA Medicines) and active ingredients (openFDA NDC). Names vary in spelling, salt form and order within combination products, so entities are identified by the name together with **brandNames**. **brandNames** and **manufacturers** occur in all three datasets and are modelled as lists, since one ingredient is sold under many trade names by many companies. Both are candidates for fusion by voting. **marketingStatus** assigns each ingredient to one of five classes (Table 4), derived from MarketingStatus (Drugs@FDA), Medicine status (EMA Medicines) and presence in openFDA NDC, which lists only currently marketed products. Because the sources cover different markets, this attribute is fused by precedence (Active > Suspended > Pending > Discontinued > Not authorised) rather than by voting. **dosageForms**, **routes** and **strengths** are shared by Drugs@FDA and openFDA NDC. **applicationNumbers** links the two FDA sources and serves only as a gold standard for evaluating identity resolution.

usApprovalDate and **euAuthorisationDate** describe distinct regulatory events and stay separate. The NDC marketing start date is not mapped, as it does not indicate approval.

Required transformations include splitting the Drugs@FDA Form field and the NDC active ingredients field (Table 3), breaking down combination products and normalising NDC application numbers.

Table 3: Attribute intersection with the integrated schema

Integrated attribute	Type	Drugs@FDA	EMA Medicines	openFDA NDC
activeIngredientName	string	ActiveIngredient	Active substance / INN	active ingredients (name) ^b
brandNames	list (string)	DrugName	Name of medicine	brand name
manufacturers	list (string)	SponsorName	MA holder	manufacturer
marketingStatus	string (categorical)	MarketingStatus	Medicine status	listing presence
dosageForms	list (string)	Form (part 1) ^a	–	dosage form
routes	list (string)	Form (part 2) ^a	–	route
strengths	list (string)	Strength	–	active ingredients (strength)
applicationNumbers	list (string)	ApplNo	–	application number
usApprovalDate	date	SubmissionStatusDate	–	–
euAuthorisationDate	date	–	Marketing authorisation date	–

^a Drugs@FDA stores dosage form and route in one field, separated by a semicolon (e.g., “TABLET;ORAL”); part 1 maps to **dosageForms**, part 2 to **routes**. Multiple routes after the semicolon are split on commas.

^b openFDA NDC stores each active ingredient as a name–strength pair; the name maps to **activeIngredientName**, the strength to **strengths**.

Table 4: Mapping of source values to `marketingStatus` classes

Class	Drugs@FDA	EMA Medicines	openFDA NDC
Active	Prescription, Over-the-counter	Authorised	listed product
Suspended	–	Suspended	–
Pending	None (Tentative Approval)	Opinion adopted, EC decision pending	–
Discontinued	Discontinued	Withdrawn, expired, revoked, lapsed	–
Not authorised	–	Refused, withdrawn application	–

2.3 Why enough entities are likely contained in multiple datasets

The three datasets collectively cover a large and overlapping portion of the global pharmaceutical market. Both Drugs@FDA and EMA Medicines focus on formally approved drugs, meaning that any substance that has received regulatory clearance in both the US and EU will appear in both sources by definition. The openFDA NDC Directory adds further breadth as a comprehensive US product listing that includes both approved and marketed drugs.

Exact normalized `name + active substance` matching currently identifies **428 shared FDA–EMA entities**, while normalized-name overlap reaches **752**. Therefore, the required $\geq 1,000$ **shared entities** have not yet been demonstrated. The third data set should, consequently, provide substantial overlap with the FDA or EMA and preferably contain active-substance information together with additional shared regulatory attributes.

As the larger of the two US sources, the openFDA NDC Directory is well positioned to fill the remaining gap. Since both FDA sources describe US-marketed products, their overlap is structurally guaranteed. Even a conservative 30% hit rate against Drugs@FDA entries would yield over 1,000 confirmed matches between the two US sources alone. Additionally, similarity-based matching will recover pairs that exact normalization misses, such as salt-form variants and biosimilar suffixes. Combining exact and approximate matching across all three sources, reaching 1,000 overlapping entities is a well-grounded expectation.

Sources

- U.S. Food and Drug Administration, *Drugs@FDA Data Files*.
<https://www.fda.gov/drugs/drug-approvals-and-databases/drugsfda-data-files>
- European Medicines Agency, *Download medicine data*.
<https://www.ema.europa.eu/en/medicines/download-medicine-data>
- U.S. Food and Drug Administration, *openFDA NDC Directory*.
<https://open.fda.gov/data/ndc/>